UNIVERSITÉ PARIS 1 PANTHÉON SORBONNE
IAE PARIS
SORBONNE BUSINESS SCHOOL

IDGP ACADEMY

Lean Six Sigma and Continuous improvement

International Certificate in Project Management and Organizational Performance

Operational efficiency is a consequence of what the customer values and how work flows — not of how busy the resources are.

Prof. Dr. Thiago Lima

International Certificate in Project Management and Organizational Performance

17 Oct 2026 · 09:00–12:00 BRT · Online



SEU PROFESSOR

PROF. DR. THIAGO LIMA

Vinte anos de prática em projetos, operações e supply chain e tomada de decisão baseada em dados

DOCÊNCIA

- Professor de MBA COPPEAD/UFRJ/IBMEC
- Coordenador da certificação em Supply Chain, Alumni COPPEAD
- Docente de mestrado, IAE Bordeaux · Professor

PRÁTICA

- Chefe do Centro de Ciências Sociais da Escola Naval
- Ex-diretor da organização responsável pela cadeia de combustíveis da Marinha
- Mais de 20 anos em logística e supply chain; contratos internacionais com multinacionais

PESQUISA E CREDENCIAIS

- Doutorado, Université de Bordeaux e Mestrado em Gestão, National University (Califórnia) ·
- Tese: riscos, estratégias, robustez e performance em cadeias de suprimentos
- Pós-doutorado no COPPEAD/UFRJ: IA aplicada à gestão hospitalar
- Lean Six Sigma Black Belt · CIAPP/IFPSM em compras · IPMA em gestão de projetos

PUBLICAÇÕES E RECONHECIMENTO

- Journal of Business & Industrial Marketing , IEEE, SBPO, SemeAd· registros de software no INPI
- Livros: Contratação e Gestão de Fornecedores Estratégicos · Excelência Operacional: da Estratégia à Execução · A Interface entre Supply Chain e Finanças Corporativas
- Prêmio Automação GS1 Brasil (2021): automação do centro de distribuição da Marinha

POR QUE ESTE SEMINÁRIO

Toda organização que adota Lean Six Sigma copia as ferramentas; poucas reproduzem o resultado. A diferença não está nas ferramentas. Esta manhã constrói a resposta em quatro equações — de onde vem o valor, o que é vantagem, como a melhoria chega ao retorno sobre o capital e por que duas operações iguais entregam resultados diferentes — e termina com o seu próprio projeto avaliado por instrumento do Professor.



Thiago Lima, PhD

Consultant & Executive Professor | Supply Chain, Projects & Data-Drive...





Thiago Lima, PhD

Consultant & Executive Professor | Supply Chain, Projects & Data-Driven Decision Making | Negotiation | Postdoctoral Researcher | Lean Six Sigma Black Belt | IPMA-D

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Thiago

wa.me/qr/WY6KFLHV7K7ME1




Education
Information
Solution
Experience
Analysis
Learning
Knowledge


DECISION MAKING
alternatives
uncertainty
high-risk consequences
interpersonal issues
complexity


SCIENCE


a better life

IAE PARIS — SORBONNE BUSINESS SCHOOL · IDGP ACADEMY

Methods, Tools and Operational Efficiency

Reading notes, instruments and programme for both sessions of the day — International Certificate in Project Management and Organizational Performance.



Prof. Dr. **Thiago Lima**

LinkedIn · Lattes CV · About the instructor

THE DAY

Saturday, 17 October 2026

Online · Brasília time · no preparation required

09:00–12:00	**Lean Six Sigma & Continuous Improvement**
14:00–17:00	**Digital Tools & Benchmarking**

Session programme

Programme (PDF)

https://tflsixsigma-png.github.io/lean-six-sigma-digital-tools-benchmarking/

THIS MORNING

Ten parts, one argument

01 The puzzle — Fremont

02 Value from the customer

03 The anatomy of waste

04 The efficiency paradox

05 Toyota, as a system

06 Lean in the language of capital

07 DMAIC

08 The third equation

09 Group activity — the Value Case

10 Where the morning ends

ONE CLAIM

Efficiency follows from what the customer values and how work flows — not from how busy the resources are. And every purchase is scaled by a capability **you cannot buy**.

WHAT THE SESSION ASKS OF YOU

Nothing to prepare — three things to have in mind

1

One recurring process

That you own or depend on, and that takes longer than it should. Contract approval, project initiation, monthly close.

2

Two rough figures

Projects in flight in your unit — and projects closed in the last twelve months. Approximations are fine.

3

One authorised project

One you could defend. One you have **private doubts about** is better material than one you are proud of.

> The examples run through factories because that is where the ideas were tested. **Queues are queues** — every block ends with a bridge to project management.

PART ONE

THE PUZZLE

Fremont, California. Same building, same equipment, substantially the same people — and **the opposite result**.

FREMONT, CALIFORNIA

Same plant. Same people. Opposite result.

1982

General Motors closes the plant

Absenteeism near **20%**. The most combative union local in GM, by GM's own description. **The worst quality** in General Motors.

1984

Reopens as NUMMI, with Toyota

Substantially the same workforce, union leadership included. Same building, same equipment. Within ~2 years: **quality and productivity lead the entire GM system**.

What changed — and where would it appear on a balance sheet?

Harder: GM saw NUMMI from the inside for two decades. Why was it never replicated?

BEFORE THE ARGUMENT STARTS · ONE LINE EACH

Write your hypothesis down

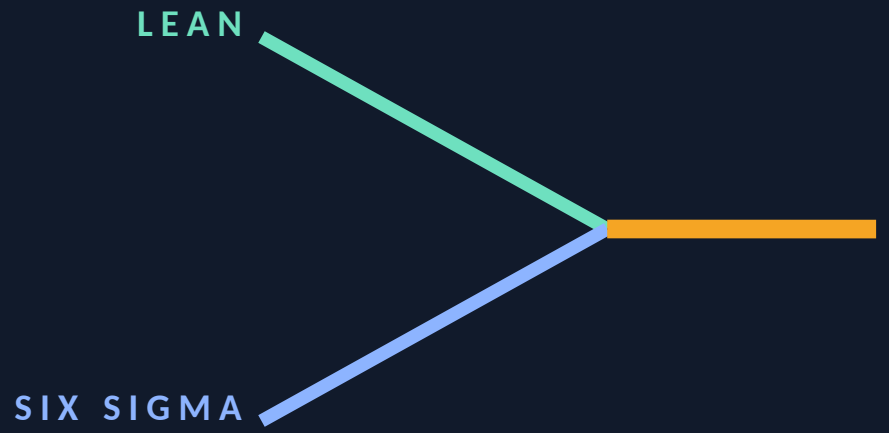
WHAT CHANGED AT FREMONT?

“What changed was _____, and it would appear on the balance sheet as _____.”

WHY COULD GM NOT COPY IT?

“GM could see _____ but could not transfer _____.”

Keep the line. **Part Eight asks you to grade it yourself** — a hypothesis nobody wrote down cannot be wrong, and it is the error that teaches.



ORIGINS

WHERE LEAN AND SIX SIGMA COME FROM

One attacks time. The other attacks variation.

A CENTURY IN ONE PICTURE

Two roots — planted the same year.

LEAN

Loom stops itself

Sakichi Toyoda: the machine halts at a defect. Jidoka is born.

Ohno builds TPS

Pull and kanban, inspired by US supermarkets.

Oil crisis exposes it

1973: Toyota recovers faster than anyone. The world notices.

"Lean" goes global

1990: The Machine That Changed the World (MIT).

SIX SIGMA

The control chart

Shewhart, Bell Labs: separate signal from noise.

Deming teaches Japan

1950: statistical quality reaches Japanese industry.

"Our quality stinks"

Motorola, 1979 → Six Sigma launched in 1986.

GE makes it a system

1995: Jack Welch turns statistics into management.

1924 — 1950s — 1970s–80s — 1990s

2002

LEAN SIX SIGMA

The two roots merge: speed and quality in one method.

THE ORIGIN OF LEAN · TOYOTA, 1945–1975

Lean was born from scarcity, not abundance.

> **Catch up with America in three years. Otherwise, the automobile industry of Japan will not survive.**
>
> **Kiichiro Toyoda,** August 1945

Toyota had no volume, no capital and no space for inventory. Copying Ford was not an option.

Sakichi Toyoda

JIDOKA

1924

A loom that stops itself when a thread breaks.

Kiichiro Toyoda

JUST-IN-TIME

1930s

Make only what is needed, when it is needed.

Taiichi Ohno

THE SYSTEM

1948–1975

Turned both into a system: pull and kanban, inspired by US supermarkets.

HOW IT TRAVELLED

Named in the West. Copied as tools.

1984

NUMMI

GM and Toyota open a joint plant in California.

1988

“Lean”

John Krafcik coins the word at MIT.

1990

MIT's landmark book

The Machine That Changed the World takes it global.

1996

Lean Thinking

Womack & Jones distil it into five principles.

THE FIVE PRINCIPLES (WOMACK & JONES, 1996)

Value → Value stream → Flow → Pull → Perfection

What travelled: the tools. What stayed behind: the management system that made them work.

THE ORIGIN OF SIX SIGMA · MOTOROLA, 1979–1986

Six Sigma was born from a quality crisis.

> **“Our quality stinks.”**
>
> **Art Sundry, Motorola executive,** 1979
>
> Japanese competitors were winning on quality in Motorola's own markets.

3.4

defects per million opportunities
The Six Sigma target (with a 1.5σ long-term shift).

1924 **Walter Shewhart, Bell Labs**

The control chart: common vs special causes.

1950 **W. Edwards Deming, Japan**

Statistical quality becomes a management idea.

1986 **Bill Smith, Motorola**

Six Sigma: a defect target for every process.

1995 **Jack Welch, GE**

From statistics to a management system.

TWO TRADITIONS, SIDE BY SIDE

Same goal. Two different enemies.

	LEAN	SIX SIGMA
BORN AT	Toyota, post-war Japan	Motorola, 1980s USA
THE ENEMY	**Waste and waiting**	**Variation and defects**
THE QUESTION	Where does the time go?	Why does the result vary?
SIGNATURE METRIC	Lead time	Defects per million (σ level)
CORE TOOLKIT	Value stream map, pull, kanban, SMED	DMAIC, control charts, DOE

Alone, each fails its own way: **Lean loses its gains; Six Sigma stays slow.**

2002 · CONVERGENCE

Speed and quality are not a trade-off.



ON MONDAY

Ask your improvement team: are we fighting time, or variation?
If nobody can answer, you are fighting neither.

PART TWO

VALUE, SEEN FROM THE CUSTOMER BACKWARDS

“All we are doing is looking at the timeline — from the moment the customer gives us an order to the point when we collect the cash.” — Taiichi Ohno

WHAT THE CUSTOMER IS ACTUALLY CHOOSING

Not every attribute buys preference

Competitive priorities

Cost · Quality · Speed · Flexibility

The operational determinants of willingness to pay. “What the customer would pay for” decomposes into these four.

Winners vs qualifiers · Hill

A **qualifier** is a threshold: its presence attracts nobody, its absence eliminates you. A **winner** makes the customer choose you once you are in the set.

Nobody picks a restaurant for its bathroom; plenty never return because it was dirty.

Eight dimensions · Garvin

Quality is not one variable. **Conformance** lowers cost; **reliability, durability, serviceability** raise willingness to pay.

The same quality effort pays twice — on two edges of value.

⚠ Effort above the threshold on a qualifier buys no preference. **A firm mid-table on every attribute has made no choice at all.**

THREE KINDS OF ACTIVITY

Where the elapsed time actually goes



<5% 10–30%

Value-adding · under 5%

Transforms the product or service in a way the customer would **knowingly pay for**.

Necessary · 10–30%

The customer would not pay for it, but **law, technology or contract** requires it.

Waste (muda)

Neither. In unimproved streams, **almost everything else is queue**.

The single most consequential number in operations — and almost no organisation measures it.

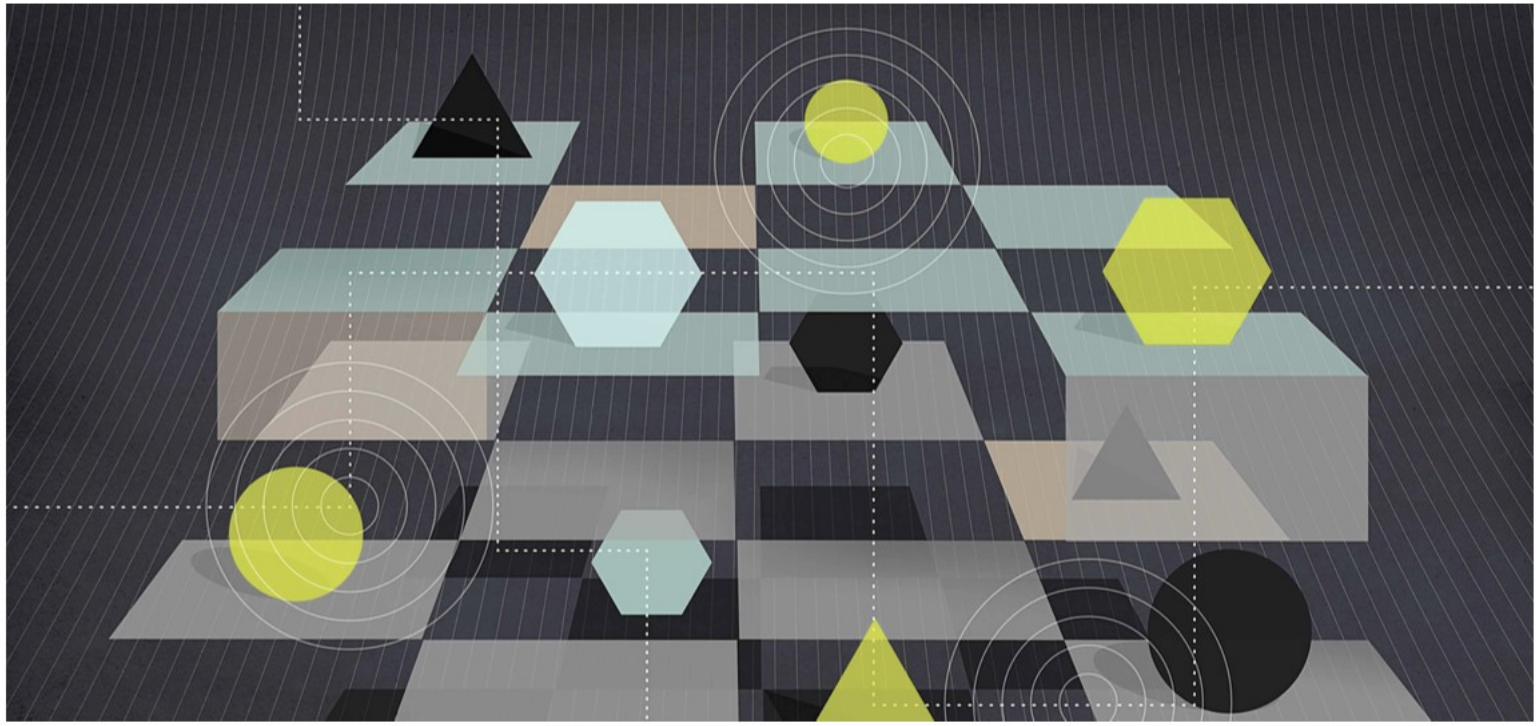
Harvard
Business
Review

Strategy

Eliminate Strategic Overload

by Felix Oberholzer-Gee

From the Magazine (May–June 2021)



Chad Hagen

IN PROJECT TERMS

If we don’t do this, do we lose...

a deal?

...or do we lose

the licence?

Winners compete. **Qualifiers get funded to the threshold.**

THE IDEA

A qualifier buys nothing above the threshold; only a winner creates preference.

IN A PROJECT

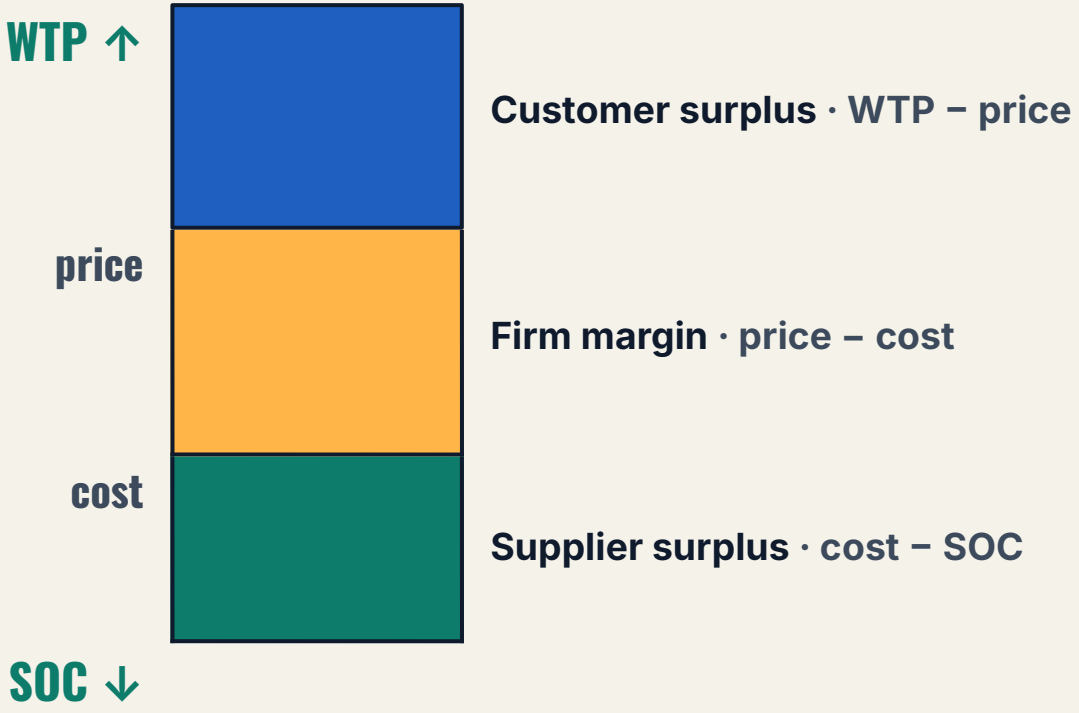
Portfolio models add strategic fit, NPV, risk and resources **on one scale** — as if every criterion were a winner. Compliance projects are qualifiers: they buy no advantage, and not doing them ends the business.

ON MONDAY

Split the list **before** scoring. Fund the “licence” column to the threshold and no further; let the “deal” column compete for what remains.

THE VALUE STICK · BRANDENBURGER & STUART, 1996

Value created = WTP − SOC



Negotiating price or supplier terms **moves** value between parties.

Only two moves **create** it: push WTP up, or push SOC down.

Lean does both at once. Removing waste lowers cost; the same work raises reliability and speed.

FROM VALUE CREATED TO ADVANTAGE

Advantage is a comparison — so write it as one

$$VC = (WTP - Cost)_{you} - (WTP - Cost)_{rival}$$

the value firm A creates, minus the value created by its relevant rival B

A rival, not last year

You cut cost 8%; the rival cuts 12%. The firm is better and **its position is worse**.

Both terms can move

VC rises when you improve — or when the rival's edge gets **harder to sustain**.

Effectiveness erodes

Visible and copyable? The rival copies, brackets converge, **VC → 0**. Porter, as arithmetic.

One test: when the competitor does this too, what is left? Nothing — fund to the threshold. Something — **name it. That residue is the advantage.**

WHERE THE RESIDUE CAN LIVE

Lima (2026), Ch. 1 — after Van Mieghem & Allon (2015)

$$V = C \times (A + P)$$

value — **capability** — **assets** — **processes**

A — Assets

Plants, machines, systems. Visible and purchasable.

P — Processes

How work flows across the assets. Imitable with effort.

C — Capability

The repeated ability to improve P. A coefficient, not a term.

WOMACK & JONES · THE FIVE PRINCIPLES

The sequence is the argument

1

Value

What is the customer paying for? From the customer backwards.

THIS PART

→

2

Value stream

Which activities deliver value, which enable it, which do neither?

PART 3

→

3

Flow

Can work move without stopping? Stoppage, not slowness, makes lead time long.

PART 4

→

4

Pull

Released by downstream demand — or pushed by upstream capacity and a plan?

PART 5

→

5

Perfection

Does the cycle repeat, the standard rewritten each time?

PARTS 7–8

Every principle after the first is meaningless until the first has been answered — and the fifth decides whether the first four survive.

PART THREE

THE ANATOMY OF WASTE

Ohno catalogued seven wastes; modern practice adds an eighth. The list matters less than its structure: **one is the root, the others are symptoms**.

THE EIGHT WASTES

In the factory — and in your projects

Waste	In a factory	In project and knowledge work
Overproduction · root	More than demand	Reports nobody reads, features nobody uses
Waiting	Idle machines and people	Approval queues, blocked dependencies
Transport	Moving materials	Handoffs between departments and systems
Over-processing	Beyond specification	Sign-off chains, redundant status meetings
Inventory	Stockpiles and WIP	Half-finished work, open tickets, unstaffed projects
Motion	Unneeded movement	Searching for information, tool switching
Defects	Scrap and rework	Rework loops, changes from unclear requirements
Unused talent	Ignored operator ideas	Experts doing admin; front-line insight ignored

Overproduction generates the other six. Attack it and they shrink; attack them one by one and it regenerates them.

THE THREE MS

Muda is the symptom. Mura and muri are the disease.

Mura · unevenness

Irregular flow creates queues — and queues create "prudent" safety buffers that are in fact **inventory**.

Muri · overburden

People and equipment beyond reasonable limits: defects, breakdowns, absence — **which create more mura**.

PRODUCE

Muda · waste

The eight wastes. The one executives learn — and **the only one most programmes attack**.

Why programmes plateau after year one: the first wave removes visible muda. Then the gains stop — the mura and muri that generate it were never touched.

IN PROJECT TERMS

120%

one specialist, across

four projects

Two is a working limit. **Four is a queue with a name.**

THE IDEA

Mura and muri are not wastes; they are what manufactures the wastes.

IN A PROJECT

A specialist at 120% across four projects produces the whole symptom set: **defects** (divided attention), **waiting** (each project queues behind three others), **unused talent** (the expert reconciles status all day).

ON MONDAY

List everyone allocated above 100%. **Count the projects per person.**

MAKING IT VISIBLE · THE VALUE STREAM MAP

A change request: four hours of work, fifteen days

Request logged	wait 2 days	Triage	wait 6 days	Technical review	wait 1 day	Approval board	wait 6 days	Implement
work 15 min		work 30 min		work 90 min		work 45 min		work 60 min

PROCESS TIME	LEAD TIME	FLOW EFFICIENCY	WHERE IT SITS
4 h	15 days	3.3%	Two approval queues

Typical flow efficiency in administrative processes: 2–10%. Every gap between current and future state needs a named change with an owner.

IN PROJECT TERMS

Two numbers on one line:

hours worked
days elapsed

The ratio is **the conversation**.

THE IDEA

The elapsed time lives in the handoffs around the work, not in the work.

IN A PROJECT

Map the administrative streams: **change request → approved change**, invoice → payment, defect reported → defect closed. They repeat, can be timed, and are where lead time accumulates.

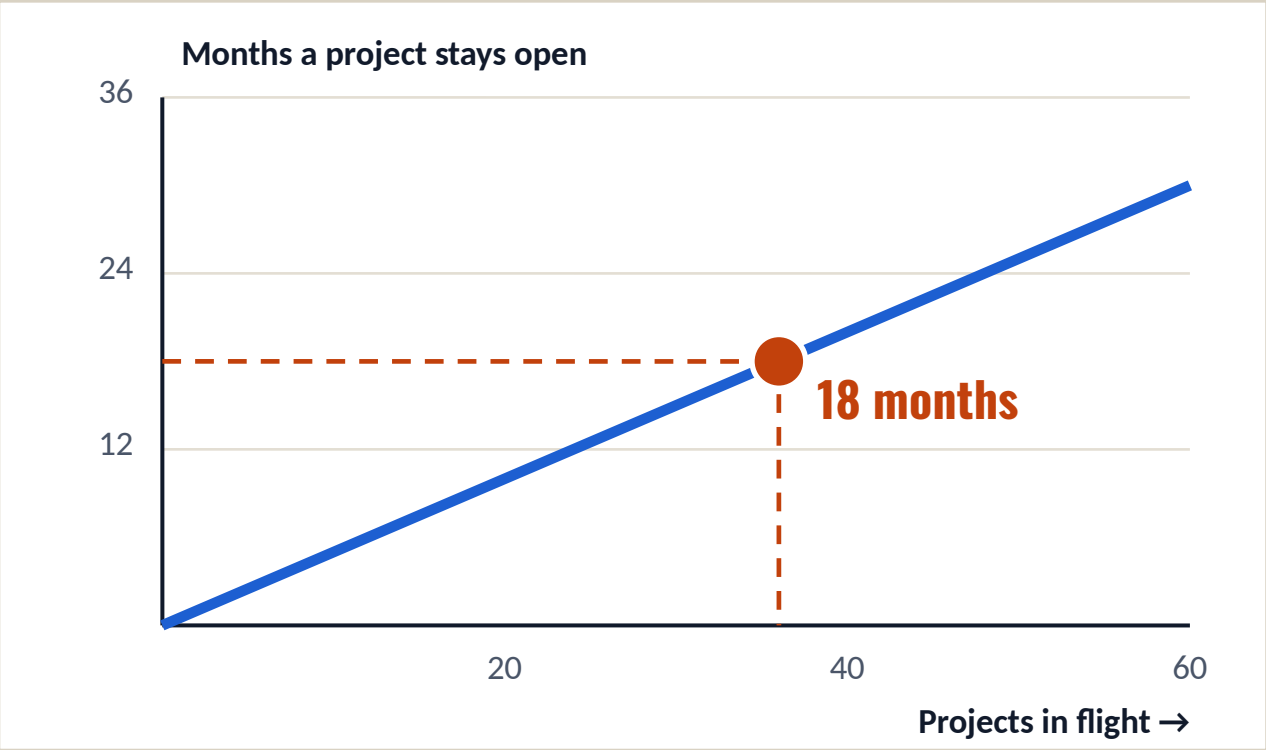
ON MONDAY

Pick one. **Walk it with the people who perform each step** — not from the procedure.

LITTLE'S LAW · AN IDENTITY, NOT A MODEL

WIP = Throughput × Lead time

Lead time ↓ → WIP ↓ → Inventory ↓ → Invested capital ↓ → ROIC ↑



FOR A PORTFOLIO

Lead time = in flight ÷ finished per year

36 in flight, 24 finished a year: **18 months** — whatever the schedules say.

Starting less finishes more.

WE DO THIS CALCULATION LIVE

Your real project lead time

A — **Projects in flight**

Everything authorised and not closed — including the untouched ones

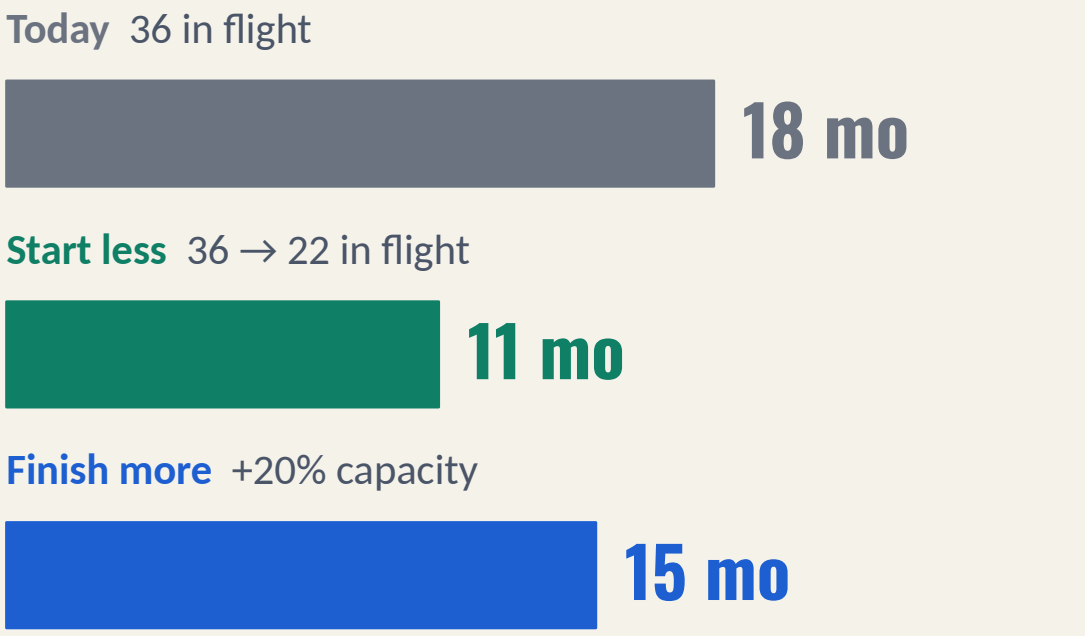
B — **Projects closed**

In the last twelve months

A ÷ B = real lead time, in years

Holds on averages, for a stable system where what starts eventually finishes.

TWO LEVERS, STARTING FROM 18 MONTHS



Start less is a decision. Finish more is a budget.

PART FOUR

THE EFFICIENCY PARADOX

Lean is routinely sold as “doing more with less”. Taken literally, **that is the definition of the thing lean exists to correct**.

RESOURCE EFFICIENCY VERSUS FLOW EFFICIENCY

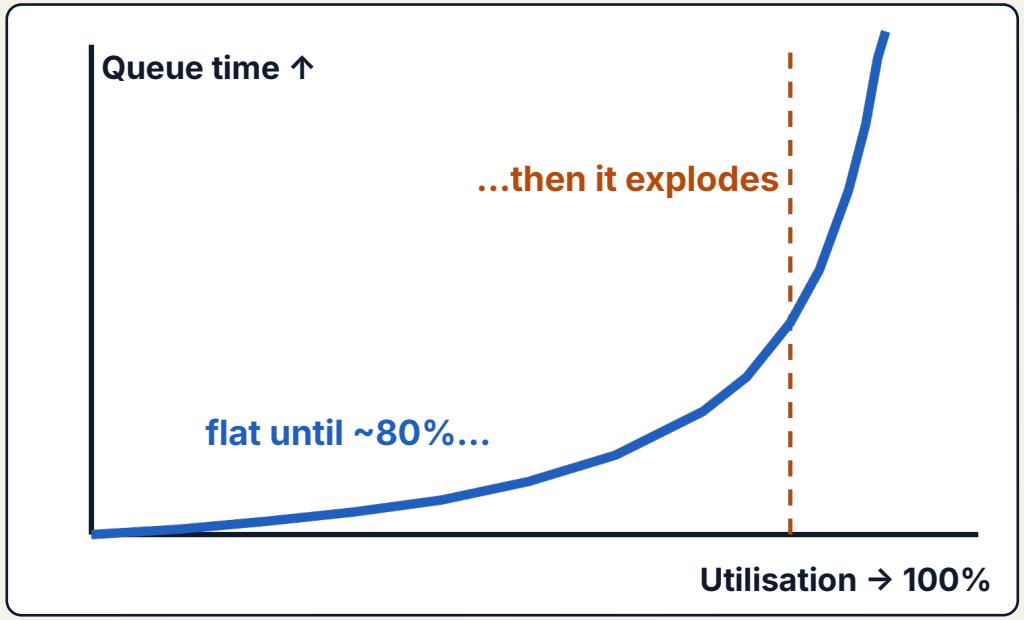
Busy resources make slow work

Resource efficiency

Keep every person and machine busy — what cost accounting measures, so organisations default to it. **Every unit looks efficient; the system queues.**

Flow efficiency

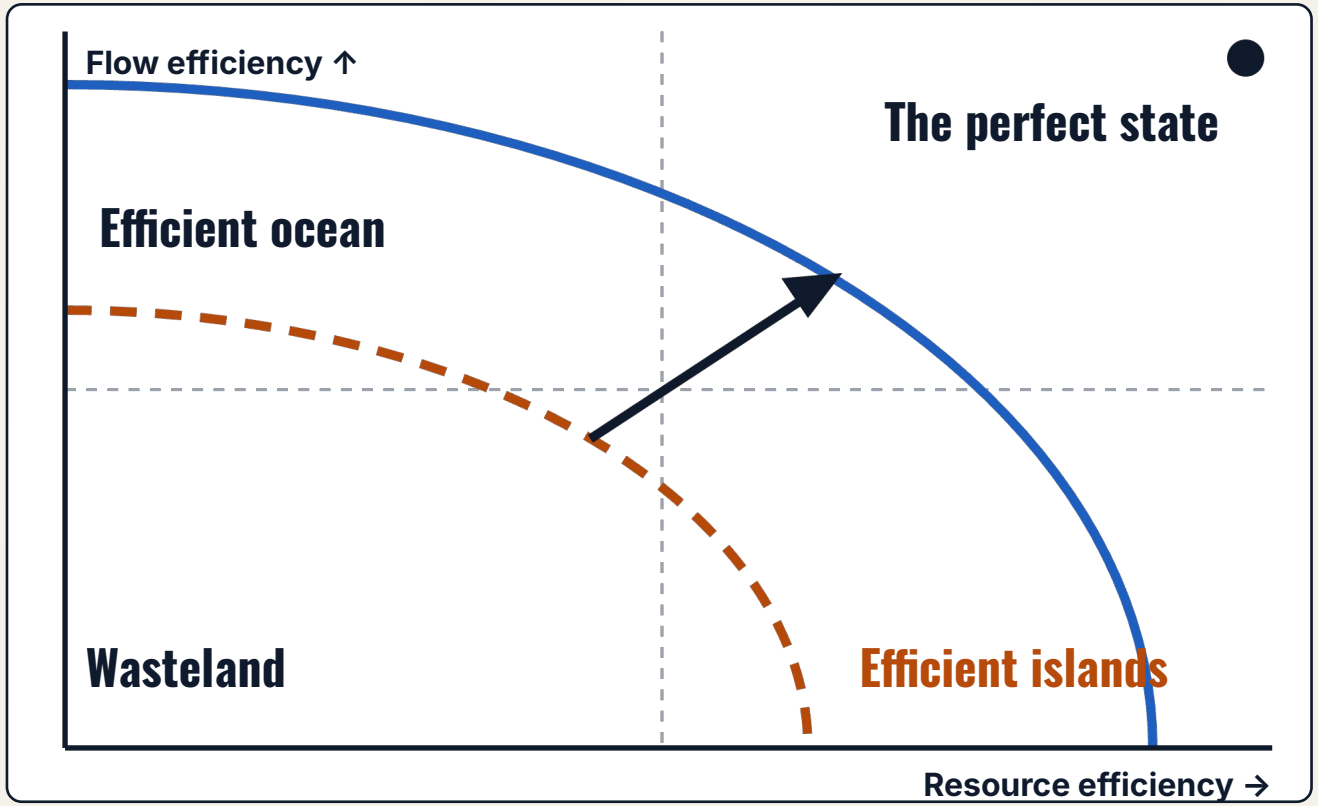
Optimise the unit's journey: **process time ÷ lead time**. Measures the order, not the resource — and almost nobody reports it.



Variability must be absorbed by **inventory**, **capacity** or **time**. Every organisation chooses one — usually without deciding. Only time is invisible in the P&L.

THE EFFICIENCY MATRIX · AFTER MODIG & ÅHLSTRÖM, 2012

Two efficiencies, one curve between them



- - - frontier today —— once variation is removed

On a frontier, pushing one efficiency up pushes the other down — **the argument never ends**.

The one move that is not a trade-off: **remove the variation** that drew the curve.

Most organisations sit bottom-right, call it productivity, and **never notice they chose a point on a poor curve**.

THE PRODUCTIVITY ILLUSION

Most companies operate in "Efficient Islands" and ignore the poor curve.

Pushing one metric up forces the other down.

THE ONLY WAY OUT

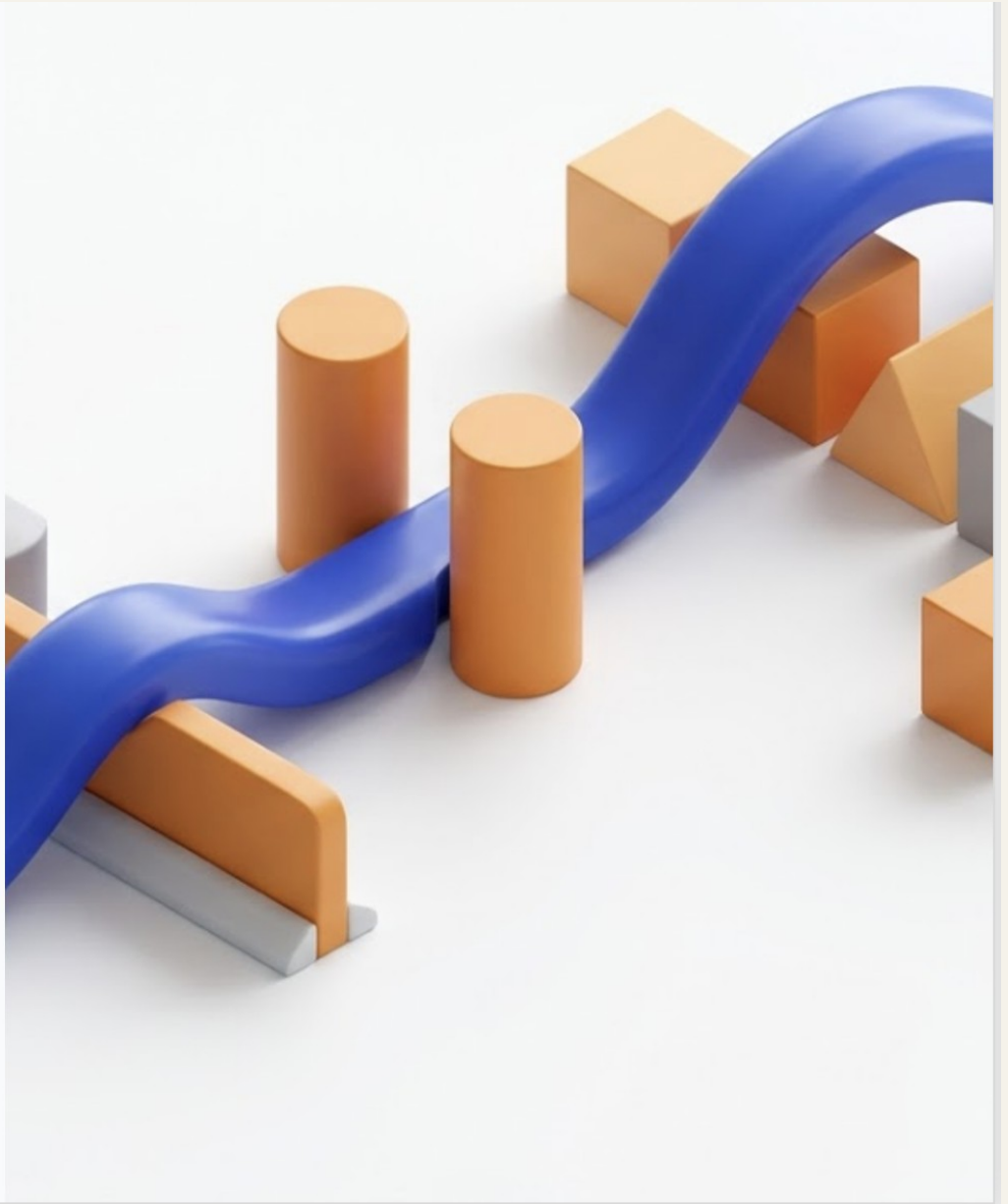
Remove Variation

End the Trade-off

Shift the frontier instead of sliding along it.

Systematic Attack

Standardize processes. Eliminate bottlenecks.



SYSTEM DYNAMICS

The End of Variation.

How to stabilize processes, protect deadlines, and break the false efficiency dilemma.

IN PROJECTS

Teams allocated at

100%

means the buffer is

the schedule.

Not choosing a buffer is choosing delay.

The Absorption Rule

The Inflexible Law

Variability will inevitably be absorbed by inventory, by capacity, or by time.

The Practical Impact

At 100% allocation, every minor illness or late approval destroys the committed deadline.

HALF THE METHOD

Neither side sustains the gain alone.

Lean without Six Sigma

Gains quickly lost.

Removes steps and initial queues. Untreated residual variation rebuilds inventory as protection.

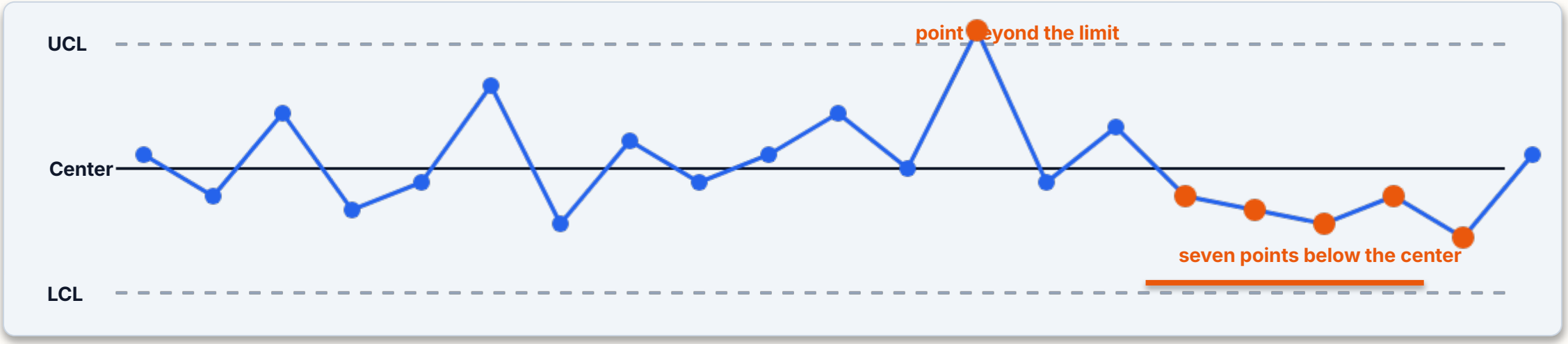
Six Sigma without Lean

Highly predictable slowness.

Perfectly capable internal steps. The overall process spends 90% of its time waiting.

SHEWHART ANALYSIS

Two signals. The rest is noise.



Common Cause

Inherent to the designed system. Investigating case by case will not reduce failure. Explaining the noise makes the process worse.

Special Cause

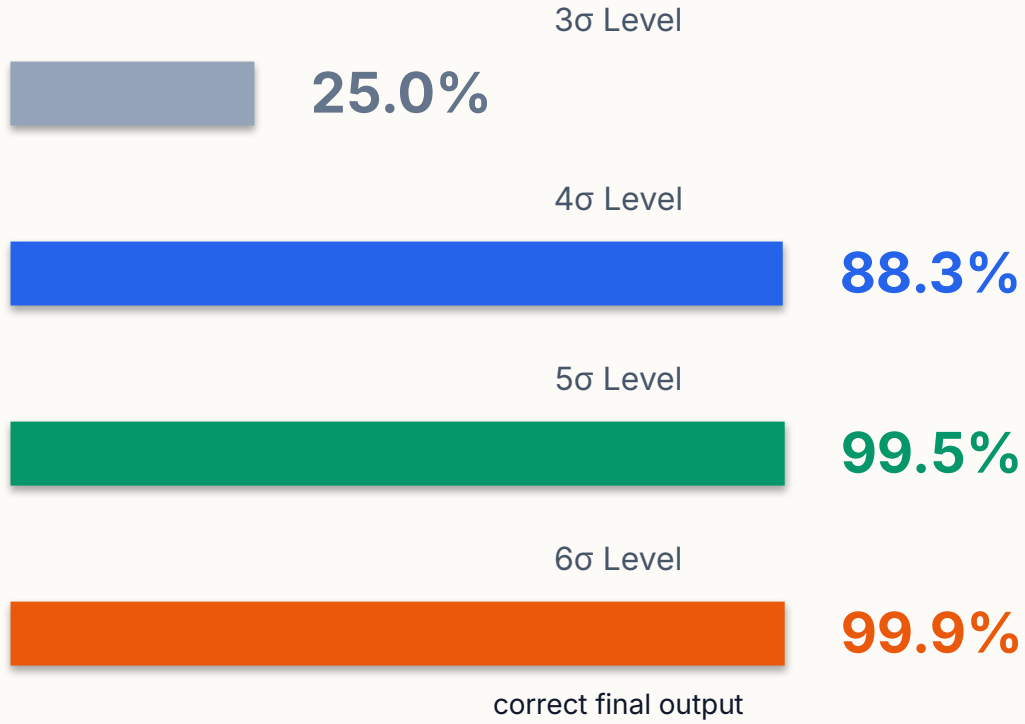
An anomaly newly introduced into the process. It can be quickly isolated and completely removed.

The Illusion of the Perfect Step

Quality is assured at the step, but experienced across the chain.

Level	Defects / Million	Step Yield
3σ	66,807	**93.3%**
4σ	6,210	**99.38%**
5σ	233	**99.977%**
6σ	3.4	**99.999%**

In a continuous 20-step process:



"Our quality is already good" is usually true at the step, but false in the final result. Every handoff eliminated is a direct quality intervention.

THE COST OF HANDOFFS

75%

defective at the end.

Even with 93,3% yield at each individual step.

This perfectly illustrates the slide's central idea that "quality is assured at the step, but experienced across the chain."

If you assess only a single step, the result looks good (nearly 94% yield).

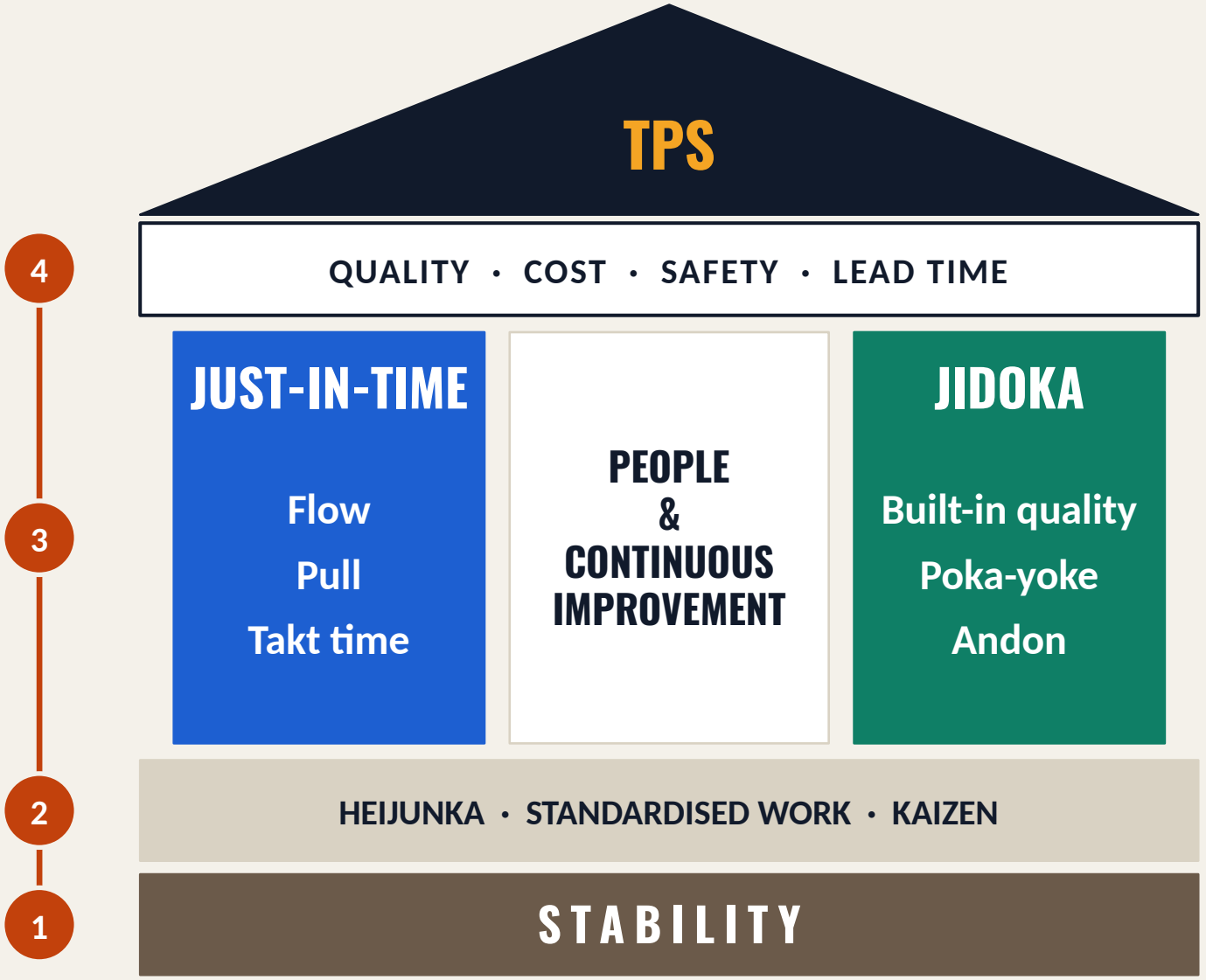
However, as small variations accumulate across 20 transfers of responsibility (handoffs), **75% of everything the process delivers at the end will have some defect** and will require rework.



PART FIVE

THE TOYOTA PRODUCTION SYSTEM, AS A SYSTEM

Remove any element and the rest collapses.



THE HOUSE

Build from the ground up.

Pull on an unstable base **starves or floods.**

Kanban on a chaotic process
= chaos, better visualised.

THE TWO PILLARS

Stop overproduction. Stop defects at the source.

JUST-IN-TIME

Make only what is pulled.

Flow	Remove the waiting
Pull	Produce on signal only
SMED	Fast changeover, small batch

Pull ≠ kanban. Pull = a cap on WIP.
Hopp & Spearman, 2004

JIDOKA

Never pass a defect on.

Poka-yoke	Make the error impossible
Andon	Signal it instantly
Stop & fix	Halt the line, fix the cause

Stop the line → problem solved. Work around it → it stays.

WHY THE SYSTEM BEATS THE TOOLS

A house, not a menu.

Partial lean can leave you worse off than no lean.

Costs without returns: the best explanation for tool-led lean failing.

MILGROM & ROBERTS, 1990

Practices pay off together, not alone.

MACDUFFIE, 1995

Bundles drive performance, not single tools.

SHAH & WARD, 2007

Lean is measured as a bundle.

ON MONDAY

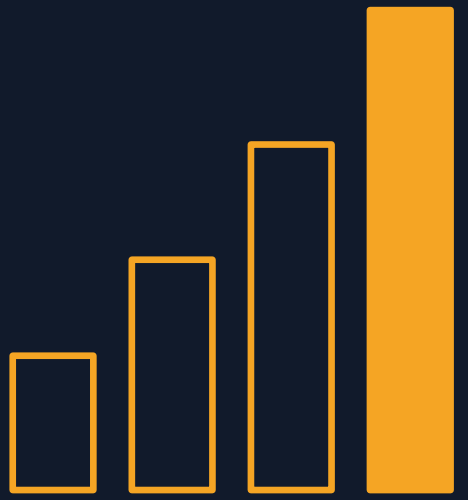
Ask what happened to the last person who reported red early.
That is your andon policy.

THE HOUSE, IN A PROJECT ORGANISATION

Most of it exists. The andon does not.

TOYOTA	IN PROJECTS	STATUS
Heijunka	Resource levelling	~ Rarely tied to a pull signal
Standardised work	Templates, PMO checklists	✓ In place
Poka-yoke	Mandatory form fields	✓ In place
Andon cord	Nothing comparable	✕ **Missing — reports do the opposite**

Why: **a red status costs the reporter more than it costs the organisation.**



PART SIX

LEAN IN THE LANGUAGE OF CAPITAL

If it can't be said in money, it won't survive the budget cycle.

WHERE LEAN SHOWS UP IN RETURN ON INVESTED CAPITAL

A cost-only case claims half of what lean does.

ROIC

= Margin × Turnover

Margin

NOPAT ÷ revenue

Defects ↓	Rework leaves operating cost
Flow ↑	Expediting stops being normal
Reliability ↑	Customers pay more

Turnover

Revenue ÷ invested capital

USUALLY MISSED

Lead time ↓	Inventory and WIP leave capital
Footprint ↓	Less space, handling, assets

Cost-only cases never count this side.

BOTH PILLARS MOVE BOTH EDGES OF VALUE

A cost-only case is undersold twice.

	LOWER COST Every business case counts this	HIGHER WILLINGNESS TO PAY Usually left out
JIDOKA	Defects removed at source: **cost of poor quality collapses.**	Zero-defect delivery lets B2B customers **drop inspection and safety stock.**
JIT	Less inventory, handling, expediting: **a smaller footprint.**	Short, reliable lead times **cut the customer's forecasting risk.**

Undersold twice: **once on willingness to pay, once on invested capital.**

What an operational improvement does to return on capital

A stylised business: revenue 1,000 a year, fixed assets 500, and each day of lead time ties up 15 of inventory. Only the two operational levers move.

TRY

Baseline

Flow only — lead time 20 → 8 days

Cost only — margin 8 → 10%

Both — the case in the text

Lead time, days	8	
Operating margin	10%	

PART SEVEN

DMAIC

Conducting improvement, analysing problems — above all, **analysing a problem before paying to solve it**.

THE FIVE PHASES

Each phase answers one question

D

Define

What is the problem, whose is it, what does it cost, where does scope end?

Charter · SIPOC · voice of the customer

M

Measure

What is today's performance, measured in a way anyone would trust?

Operational definitions · MSA

A

Analyze

What is the cause — **verified, not assumed**?

Five whys · Ishikawa · Pareto · hypothesis tests

I

Improve

Does the solution work in a pilot, before it is scaled?

Pilot · poka-yoke · DOE

C

Control

Does the gain survive the dissolution of the project team?

Control charts · control plan · standard work

Define and Measure are half the work. A badly defined problem has no verifiable solution; an untrusted measure ends in opinion.

Not rival systems. Poka-yoke appears here and in Jidoka. Lean fights waste; Six Sigma fights variation.

THE FAILURE MODE IT EXISTS TO PREVENT

Not statistics. Discipline.

WHAT EVERYONE REMEMBERS

3.4 per million

Motorola, the 1980s: limits six standard deviations from the mean. **Not the part that matters most** to a senior manager.

WHAT MATTERS

Pattern-matching

The experienced manager recognises the problem in seconds — fast, often right, **entirely unverified**. When wrong, the error is found after payment, and teaches nothing.

DMAIC does not distrust experience. It makes the hypothesis **written down before it is funded** — so it can be refuted while refutation is still cheap.

ANALYZE, IN PRACTICE

How to ask why — and where initiatives die

Five whys — two rules

1 · A person is not a process cause. “He didn’t deliver on time” restates the symptom. Ask: what in the process lets that happen unnoticed?

2 · Stop at something somebody controls. “The market is volatile” went one step too far; the second link is not far enough.

Where initiatives die

In Analyze: the root cause turns out to be **whatever someone already wanted to buy**.

In Control: the gain lasts as long as the team. Then it regresses — **nobody inherited the standard**.

The cheapest safeguard: a tollgate per phase, chaired by the sponsor who owns the result. Whoever conducts the work is not whoever approves it.

IN PROJECT TERMS

At the next approval, ask:

“What is the verified cause?”

Not for the exploratory front end: there, DMAIC suppresses the variation discovery needs. **Two regimes.**

THE IDEA

A conventional plan commits to a solution at authorisation; DMAIC forbids naming one until a cause is verified.

IN A PROJECT

Charter, tollgates, sponsor — the vocabulary overlaps. **The Analyze → Improve gate** has no equivalent in most governance, and it does the work. Control is benefits realisation: described everywhere, performed rarely.

ON MONDAY

If the answer is a solution — “a new system”, “more people” — **the project is at Define**, whatever gate it is passing.

PART EIGHT

THE THIRD EQUATION

Stated in Part Two before there was anything to test it against. Everything since has been that test. **Now the puzzle gets its answer.**

THE THIRD EQUATION

V = C × (A + P)

Term	What it is	Visibility
A — Assets	Plants, machines, systems. What the pillars stand on.	Visible, purchasable
P — Processes	How work flows. The pillars and the foundation.	Partly visible, imitable
C — Capability	The repeated ability to improve P. The centre of the house.	Barely visible, rarely imitable

THREE QUESTIONS FOR THE PMO

What do we deliver better this year — and how do we know?

Which improvement survived its author leaving?

What would remain if the tool were replaced tomorrow?

You can buy a plant, license a platform, redraw a process. **You cannot buy the repeated ability to improve — and if C → 0, V → 0.**

THE PUZZLE, ANSWERED

What changed at Fremont was C.

What changed — and where does it appear?

Not A: same building, same equipment. Not only P: GM copied the processes for twenty years. It was **the repeated ability to find and fix its own problems** — with the same people. On the balance sheet: **nowhere**. Only in its consequences.

Why was it never replicated?

GM could observe **the bracket**: kanban, andon, standard work sheets. What produced the result was **the coefficient** — rules-in-use nobody at Toyota needed to write down.

✓ **Now grade the line you wrote at 09:10. Did it name the bracket, or the coefficient?**

THE EQUATIONS, IN ONE PLACE

Four equations, one argument

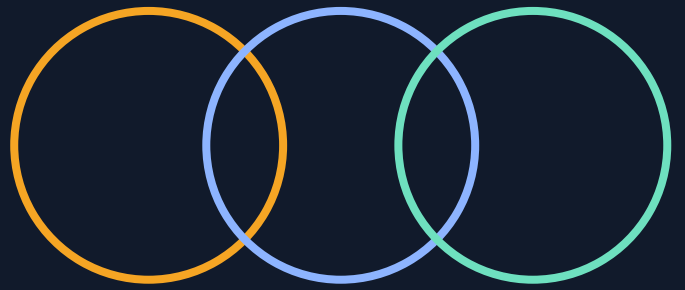
Value stick · WTP − SOC	Lean pushes both edges at once: Jidoka and JIT each lower cost and raise WTP.
VC = (WTP − C)you − (WTP − C)rival	Creating more value than the rival is what the firm is paid for. Copyable gains return VC to zero.
ROIC vs WACC	Lean lifts the numerator and cuts the denominator — return above the cost of capital.
V = C × (A + P)	**…but only for organisations that build the multiplier rather than buy the bracket.**

PORTER'S OBJECTION

Operational effectiveness is imitable, so its gains compete away. Where must strategy-grade advantage live?

SPEAR & BOWEN'S FINDING

Toyota's edge is its tacit rules-in-use, not its tools. Tacit rules can be observed and still not transferred.



PART NINE · GROUP ACTIVITY

THE VALUE CASE

One real project. Three lenses: willingness to pay · waste · what it builds.

WHY THIS IS NOT A BUSINESS CASE

Three questions a business case never asks.

1
Which edge of value?

Every rival competes on cost. **Raising willingness to pay is rarer — and under-argued.**

2
A named waste, or an aspiration?

“+15% efficiency” is a target. **“Eliminates three re-approvals” can be verified.**

3
More capable, or better equipped?

Rivals can buy the same A and P. **Getting better at improving cannot be bought.**

90
SECONDS

Choose one authorised project a member can defend.

Pick the one the group disagrees about — not the best business case.

THE VALUE CASE CANVAS

Six steps, fifteen minutes.

PRESENTED

1 The project

What it is, who sponsors it, what it was approved to achieve.

2 The WTP edge

Would the customer notice? Winner or qualifier?

3 Map the stream

3–6 steps. Process time vs lead time: where is the gap?

4 The cost edge

Name the waste — not a benefit. Where, and how big?

5 What it builds

Split into A, P and C. Training on the new system = delivering the asset.

PRESENTED

6 The verdict

How long to copy? Fund · fund with conditions · do not fund.

One spokesperson presents steps 1 and 6. **Export the canvas before closing the tab.**

WHAT TO EXPECT

Two findings most groups will reach.

“Only one edge moves.”

Most portfolios are almost all cost-edge projects. Not wrong — but an unexamined choice.

ASK **What would it take to also move WTP?**

“C is empty.”

A and P get delivered because they can be specified, scheduled and signed off. C resists all three.

ASK **What in this portfolio builds advantage?**

Neither is a failure — both are findings. Bring the export to the afternoon: it feeds the Execution Optimization Challenge.



PART TEN

WHERE THE MORNING ENDS

Five moves · four equations · three things that should survive six months.

THE ARGUMENT

The morning in five moves.

1

Value starts at the customer

Winners create preference; qualifiers only avoid elimination.

2

Waste is the gap

Hours of work inside weeks of elapsed time — almost all waiting.

3

Utilisation creates waiting

Busy people don't cure it. Customers feel flow efficiency.

4

Variation is the other half

Lean removes waiting; Six Sigma narrows spread. Neither holds alone.

5

It all shows up in ROIC

In the margin — and in the denominator business cases forget.

FOUR EQUATIONS, ONE STAIRCASE

Each step answers what the last one left open.

LEAN
moves both edges: waste out lowers cost; speed and reliability raise WTP

1 · VALUE

WTP − SOC

Where value comes from — and the only two moves that create it.

SIX SIGMA
holds the gain: fewer defects lift the margin, less inventory shrinks the capital

2 · ADVANTAGE

VC = you − rival

Creating value is not advantage. A copyable gain returns it to zero.

DMAIC, REPEATED
becomes C: the next cycle runs without this team

3 · RETURNS

ROIC = margin × turnover

How improvement reaches those who fund it — both levers at once.

4 · CAPABILITY

V = C × (A + P)

Same assets and processes, different results: where advantage lives.

LEAN SIX SIGMA is not a fifth step — it is the method that climbs all four.

WHAT THE AFTERNOON TESTS

When you can't build C, you buy it. Two ways.

1 Technology

The most expensive item — and the easiest to mistake for capability. **AI is the extreme case.**

2 Other people's practices

Benchmarking looks like it should work. GM toured NUMMI for twenty years and **copied everything that could be photographed.**

14:00
AFTERNOON

MODULE 3 · SESSION 4 · MORNING

THANK YOU

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Reading notes, simulators and the Value Case Canvas: tflsixsigma-png.github.io/lean-six-sigma-digital-tools-benchmarking

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